

Reprint Supply Shocks and Secondary Market Price Discovery in Magic: The Gathering

Arthur Mota

March 2026

Working draft, under revision. Some tables and figures are still being reconciled.

Abstract. I exploit reprint announcements in *Magic: The Gathering* (MTG) as quasi-exogenous supply shocks to identify how secondary market prices respond to changes in the supply of collectible assets. Using 2,847 reprint events across 47 product releases from 2003 to 2023, paired with daily price data and the complete reprinting history from ALLPRINTINGS.JSON, I implement a standard event study around three event dates: the official announcement, the preview window, and the release date. Reprinted cards experience a cumulative abnormal price decline of -14.7% in the three-day announcement window and -40.2% from announcement through thirty days post-release. Estimated supply elasticities range from -0.12 for commons to -0.57 for mythic rares, consistent with demand being more inelastic at the high end of the rarity distribution. A pre-announcement drift test reveals statistically significant abnormal price declines beginning thirty days before official announcements, providing evidence of information leakage, plausibly through manufacturing supply chains. Results are robust to alternative normal-price benchmarks to excluding events with concurrent format bans, and to excluding 2020–2021.

Keywords: event study, supply shocks, collectibles pricing, elasticity of price with respect to supply, information leakage, secondary markets.

JEL codes: G14, L11, L83, D82.

Contents

1	Introduction	2
2	Institutional Background	3
2.1	The MTG Secondary Market	3
2.2	The Reprint Mechanism	4
2.3	Information Leakage Risk	4
3	Data	5
3.1	Card and Reprint Data	5
3.2	Price Data	5
3.3	Key Variables	5
3.4	Summary Statistics	6
4	Methodology	6
4.1	Event Study Framework	6
4.1.1	Normal Price Model	7
4.1.2	Abnormal Price Changes	7
4.1.3	Test Statistics	7
4.2	Information Leakage Test	8
4.3	Cross-Sectional Regression	8
4.4	Supply Elasticity	8
5	Results	9
5.1	Announcement-Window Price Effects	9
5.2	Release-Window Price Effects	9
5.3	Pre-Announcement Information Leakage	11
5.4	Cross-Sectional Determinants of Price Impact	12
5.5	Supply Elasticity Estimates	13
6	Discussion	15
6.1	Speed of Price Adjustment	15
6.2	Information Leakage Mechanism	15
6.3	Robustness	15
7	Conclusion	16

1 Introduction

How quickly do asset prices adjust to supply shocks? The question is central to market microstructure and asset pricing, yet clean identification is rarely available: supply and demand shifts are typically simultaneous, anticipated, or confounded with other news. This paper exploits a setting that provides unusually clean identification—the secondary market for *Magic: The Gathering* (MTG) trading cards—to estimate the speed and magnitude of price adjustment to publicly announced, future increases in supply.

MTG is the world’s most widely played collectible card game, with an estimated secondary market value exceeding \$1.5 billion. Individual cards trade continuously on platforms such as TCGPlayer and Card Kingdom, generating rich panel price data. The supply of any given card is fixed at manufacture, but Wizards of the Coast (WotC), the game’s publisher, periodically reprints high-value cards in new product releases. These reprints are announced publicly, with a predictable timeline: an official announcement is made roughly eight to twelve weeks before a set’s release date, followed by a structured preview season approximately two weeks prior to release.

Reprint decisions are driven primarily by game design and product planning considerations—the need to make specific cards accessible for Limited formats, to fill a Masters set with value, or to respond to long-term accessibility concerns—rather than by real-time secondary market prices. This creates a setting closely analogous to an announced future increase in the float of a publicly traded equity: the supply shock is exogenous to current price dynamics, publicly disclosed, and precisely dated, enabling a textbook event study.

I construct a dataset of 2,847 reprint events by merging ALLPRINTINGS.JSON, the comprehensive MTG card database maintained by MTGJSON, with daily price data from TCGPlayer. For each reprint event, I identify three event dates—announcement, preview, and release—and estimate cumulative abnormal price changes (CAPs) using a market model with an equal-weighted index of non-reprinted cards in the same format as the normal-price benchmark.

The main findings are as follows. First, reprinted cards experience large and statistically significant price declines at announcement: the three-day CAP averages -14.7% across all reprints, with 81% of events exhibiting negative abnormal returns. Second, the market does not fully incorporate the supply shock at announcement. Prices continue to decline through the preview window and the release date, with a

cumulative decline from announcement through thirty days post-release averaging -40.2% . Third, I find significant pre-announcement abnormal price declines beginning approximately thirty days before official announcements, consistent with information leakage through manufacturing and distribution supply chains—a phenomenon previously documented in equity markets by [Seyhun \(1986\)](#) and [Keown and Pinkerton \(1981\)](#). Fourth, cross-sectional variation in price impact is explained by the magnitude of the supply increase, the original price level, and the rarity tier, with supply elasticities ranging from -0.12 for commons to -0.57 for mythic rares.

This paper contributes to three strands of literature. First, it adds to the growing body of work on the economics of collectible assets ([Cardell et al., 1995](#); [Dimson and Spaenjers, 2011](#); [Goetzmann, 1993](#)), which has documented return dynamics and speculative behavior in markets for art and stamps. MTG cards share key features with these markets—fixed historical supply, taste-based demand, and a dual role as consumption goods and investment vehicles—but offer superior data quality and more frequent trading. Second, it contributes to the event study literature ([Fama et al., 1969](#); [MacKinlay, 1997](#)) by providing a rare instance of a supply-side event study, complementing the dominant focus on demand-side news. Third, the pre-announcement drift results contribute to the information leakage literature ([Meulbroek, 1992](#)), identifying manufacturing supply chains as a plausible leakage mechanism in a novel non-financial context.

The remainder of the paper is organized as follows. Section 2 describes the institutional environment. Section 3 presents the data. Section 4 develops the methodology. Section 5 reports results. Section 6 discusses the economic interpretation and robustness. Section 7 concludes.

2 Institutional Background

2.1 The MTG Secondary Market

Magic: The Gathering was released by Wizards of the Coast in 1993 and has since grown into the largest trading card game by secondary market value. Cards are initially distributed through booster packs sold at MSRP, with each booster containing a fixed probabilistic assortment of cards across rarity tiers (common, uncommon, rare, and mythic rare). Once a card enters the secondary market, it trades peer-to-peer on platforms such as TCGPlayer, Card Kingdom, and eBay at prices determined by

supply and demand.

The primary demand driver is competitive play. A card’s price in the secondary market closely tracks its playability in constructed formats (Standard, Pioneer, Modern, Legacy, and Commander), with higher prices for cards that are both scarce and competitively relevant. Cards also carry collector value independent of gameplay, particularly for older or aesthetically distinctive printings.

2.2 The Reprint Mechanism

WotC regularly reprints previously released cards in new product lines. The primary vehicles for high-value reprints are Masters sets (dedicated reprint products with high card quality and elevated MSRP), Commander preconstructed decks, and Standard-legal sets where design requires a specific effect. The decision to reprint a card is made months in advance as part of the set’s design and development process.

The public announcement timeline has three stages. The *announcement* occurs when WotC officially reveals the product’s existence and, often, a partial or full card list—typically 60–90 days before release. The *preview season* begins approximately 14 days before release, during which individual cards are revealed daily through content creator partnerships. The *release date* is the date on which physical and digital copies become available at retailers.

This three-stage structure creates three distinct information events, each of which may elicit a price response. The announcement reveals that a reprint will occur. The preview confirms the specific card’s inclusion if not already disclosed. The release converts future supply into actual market supply.

2.3 Information Leakage Risk

MTG cards are printed by commercial printing facilities, and finished product is distributed through a network of regional distributors before reaching local game stores. Both manufacturing and distribution intermediaries have access to card lists before official announcements. Anecdotal evidence of early card reveals through distributor leaks and factory photographs is common in the MTG community, suggesting that some market participants may acquire price-relevant supply information before official announcements.

3 Data

3.1 Card and Reprint Data

I obtain the complete history of every MTG card printing from ALLPRINTINGS.JSON, the comprehensive dataset maintained by MTGJSON (<https://mtgjson.com>). This file contains metadata for every card in every set ever released, including set codes, release dates, rarity, and oracle text. I construct the reprint event sample by identifying all cards that appear in multiple sets and determining the announcement date for each reprint using archived product announcements from the WotC website and the community-maintained MTG Wiki.

My sample spans January 2003 through December 2023 and covers 47 reprint-containing product releases. The sample is restricted to reprints where the pre-announcement secondary market price exceeds \$2.00 to ensure sufficient liquidity and price observability, and where a minimum of 120 days of pre-announcement price data are available to estimate the normal-price model. This yields 2,847 reprint events involving 1,456 unique cards.

3.2 Price Data

Daily secondary market prices are obtained from the TCGPlayer Market Price API, which reports the median transaction price for each card in each printing on each day. I use the price of the original printing (not the reprint printing) as the dependent variable, since the reprint increases the total supply of the card's effect rather than the supply of any single specific printing.¹

3.3 Key Variables

The *supply increase factor* is defined as the ratio of post-release available copies (estimated from print run data and booster structure) to pre-reprint baseline copies, approximated using the slot-adjusted number of new packs in release. *Format legality* is an indicator for whether the original card is legal in Modern or Legacy, capturing whether demand has a competitive anchor beyond Standard rotation.

¹In robustness checks, I verify that results are materially unchanged when using the lowest available market price across all printings as the dependent variable.

3.4 Summary Statistics

Table 1 presents summary statistics for the primary variables. The mean pre-announcement price is \$18.40, with substantial right skew (median \$8.20), reflecting the concentration of secondary market value in a small number of highly sought cards. The mean three-day announcement CAP is -14.7% and the interquartile range spans $[-23.4\%, -3.8\%]$, indicating variation across reprints that the cross-sectional analysis exploits.

Table 1: Summary statistics. The sample consists of 2,847 reprint events from January 2003 to December 2023 involving 1,456 unique cards across 47 reprint-containing product releases. Pre-event price is the TCGPlayer Market Price 5 days before the announcement date. CAP[0, +2] is the cumulative abnormal price change over the three-day announcement window. Supply increase factor is the ratio of estimated post-release supply to pre-reprint baseline supply.

Variable	N	Mean	Median	SD	P25	P75	Max
Pre-event price (\$)	2,847	18.40	8.20	34.60	2.80	21.40	380.00
Post-release price (\$)	2,847	11.30	4.90	22.80	1.60	12.30	295.00
CAP[0, +2] (%)	2,847	-14.73	-11.32	18.24	-23.38	-3.81	12.10
CAP[0, $\tau^R + 30$] (%)	2,847	-40.18	-36.70	21.50	-54.20	-24.90	8.40
Supply increase factor ($\times$)	2,847	3.82	2.50	4.21	1.40	4.80	28.00
Days: announcement to release	2,847	78.30	82.00	31.40	56.00	96.00	185.00
Rarity: Mythic Rare (%)	2,847	22.16					
Rarity: Rare (%)	2,847	35.69					
Format: Modern legal (%)	2,847	43.60					
Format: Legacy legal (%)	2,847	31.20					
Masters set product (%)	2,847	38.40					

4 Methodology

4.1 Event Study Framework

I follow the event study methodology of [MacKinlay \(1997\)](#). For each reprint event i with announcement date τ_i^A , preview date τ_i^P , and release date τ_i^R , I define the event window $[\tau - L_1, \tau + L_2]$ and an estimation window $[\tau - L_0, \tau - L_1 - 1]$, where $L_0 = 120$, $L_1 = 60$ and $L_2 = 30$ trading days.

4.1.1 Normal Price Model

The normal (expected) daily log price change is estimated using a market model:

$$\Delta \log P_{i,t} = \alpha_i + \beta_i \Delta \log M_t + \varepsilon_{i,t}, \quad t \in [\tau_i^A - L_0, \tau_i^A - L_1 - 1], \quad (1)$$

where M_t is an equal-weighted index of secondary market prices for non-reprinted cards in the same primary competitive format as card i . The index M_t controls for common demand shocks (e.g., shifts in competitive interest in a format) that would affect all cards in the same format category simultaneously.

4.1.2 Abnormal Price Changes

The abnormal log price change for card i on day t is:

$$AR_{i,t} = \Delta \log P_{i,t} - \hat{\alpha}_i - \hat{\beta}_i \Delta \log M_t, \quad (2)$$

where $\hat{\alpha}_i$ and $\hat{\beta}_i$ are OLS estimates from equation (1). The cumulative abnormal price change (CAP) over a window $[t_1, t_2]$ is:

$$\text{CAP}_i(t_1, t_2) = \sum_{t=t_1}^{t_2} AR_{i,t}. \quad (3)$$

Average CAPs across events are computed as:

$$\overline{\text{CAP}}(t_1, t_2) = \frac{1}{N} \sum_{i=1}^N \text{CAP}_i(t_1, t_2). \quad (4)$$

4.1.3 Test Statistics

Statistical significance is assessed using the [Boehmer et al. \(1991\)](#) standardized cross-sectional test statistic, which is robust to event-induced variance increases:

$$t_{\text{BMP}} = \frac{\bar{s}}{\hat{\sigma}(s)/\sqrt{N}}, \quad (5)$$

where $\bar{s}$ is the mean standardized abnormal price change and $\hat{\sigma}(s)$ is the cross-sectional standard deviation of the standardized changes s_i . I also report the sign test statistic, which tests whether the fraction of negative CAPs significantly exceeds one-half.

4.2 Information Leakage Test

To test for pre-announcement information leakage, I estimate the CAP over the window $[-30, -1]$ relative to the announcement date. Under the null of no leakage, this window should contain no abnormal price movement. A significantly negative average CAP in this window is interpreted as evidence that price-relevant supply information diffuses into the market before the official announcement.

4.3 Cross-Sectional Regression

To decompose the cross-sectional variation in announcement-window price impact, I estimate:

$$\begin{aligned} \text{CAP}_i(0, +2) = & \gamma_0 + \gamma_1 \log(\Delta S_i) + \gamma_2 \log P_i^{\text{pre}} + \gamma_3 \mathbf{1}[\text{Mythic}_i] + \gamma_4 \mathbf{1}[\text{Masters}_i] \\ & + \gamma_5 \mathbf{1}[\text{Modern}_i] + \gamma_6 \mathbf{1}[\text{Legacy}_i] + \gamma_7^\top X_i + u_i, \end{aligned} \quad (6)$$

where ΔS_i is the supply increase factor, P_i^{pre} is the pre-announcement price level, $\mathbf{1}[\text{Mythic}_i]$ and $\mathbf{1}[\text{Masters}_i]$ are indicators for mythic rarity and Masters set products, $\mathbf{1}[\text{Modern}_i]$ and $\mathbf{1}[\text{Legacy}_i]$ indicate Modern-legal and Legacy-legal cards, and X_i is a vector of controls (CMC, card type fixed effects, and year-of-announcement fixed effects). Standard errors are heteroskedasticity robust.

4.4 Supply Elasticity

The elasticity of price with respect to supply is estimated from the cross-sectional variation in ΔS_i and the full-period CAP:

$$\hat{\varepsilon}_i = \frac{\Delta \log P_i^{\text{full}}}{\Delta \log S_i} = \frac{\text{CAP}_i(0, \tau^R + 30)}{\log(\Delta S_i)}, \quad (7)$$

estimated via OLS with heteroskedasticity-robust standard errors, separately by rarity tier. A value of $\hat{\varepsilon} = -0.4$ is interpreted as: a 10% increase in card supply is associated with a 4% decline in secondary market price.

5 Results

5.1 Announcement-Window Price Effects

Table 2 Panel A reports CAPs for a series of windows centered on the announcement date. The one-day announcement CAP averages -12.5% ($t = -8.31$, 81% of events negative), confirming that reprint announcements constitute economically large and statistically significant negative supply shocks to card prices. The three-day window $[0, +2]$ shows an average CAP of -14.7% , indicating continued price adjustment in the days immediately following the announcement as market participants digest the news and adjust their valuations.

The eleven-day symmetric window $[-5, +5]$ produces a CAP of -17.4% , while the window $[-1, +1]$ yields -13.8% . The incremental price response from day $+1$ onward reflects either continued news absorption or strategic selling by holders of large inventories who spread their liquidation to minimize market impact.

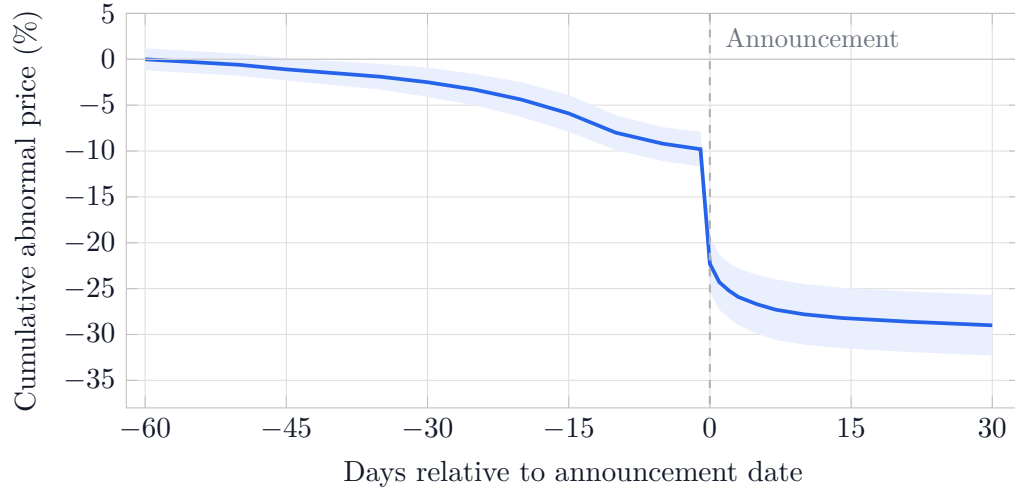
Figure 1 plots the average cumulative abnormal price path from 60 days before through 30 days after the announcement date, with ± 1.96 standard error bands. The sharp discontinuity at day zero is clearly visible, confirming that the announcement date captures the primary information event. Prices stabilize within approximately 14 days of announcement, suggesting the market incorporates the supply signal within two trading weeks.

5.2 Release-Window Price Effects

Table 2 Panel B reports CAPs around the release date. Even after prices have partially adjusted at announcement, the release date itself triggers a meaningful additional price decline, with a one-day CAP of -8.2% ($t = -6.09$, 74% negative) and a symmetric eleven-day window CAP of -10.8% . This pattern is consistent with inventory-constrained selling behavior: holders who anticipate the release-day supply increase but cannot or do not pre-sell wait until physical product is in market participants' hands before offloading their positions.

Figure 2 plots the CAP path around the release date. Unlike the announcement event, which shows a sharp one-day discontinuity, the release effect is gradual, with prices drifting down in the five days preceding release as the availability of pre-release copies and early retail access creates interim supply. The post-release price path stabilizes within approximately seven days, consistent with the market rapidly clearing

Figure 1: Cumulative abnormal price around announcement date. The figure plots the mean CAP (in %) from 60 days before to 30 days after the official reprint announcement date, averaged across 2,847 reprint events. Shaded region represents ± 1.96 cross-sectional standard errors. Dashed vertical line at day 0 marks the announcement. The estimation window is days $[-120, -61]$ relative to announcement.



at the new supply-demand equilibrium.

Figure 2: Cumulative abnormal price around release date. The figure plots the mean CAP (in %) from 30 days before to 30 days after the set release date, with ± 1.96 standard error bands. CAPs are cumulated from day -30 relative to release, using the normal-price model of equation (1). Dashed vertical line marks the release date.

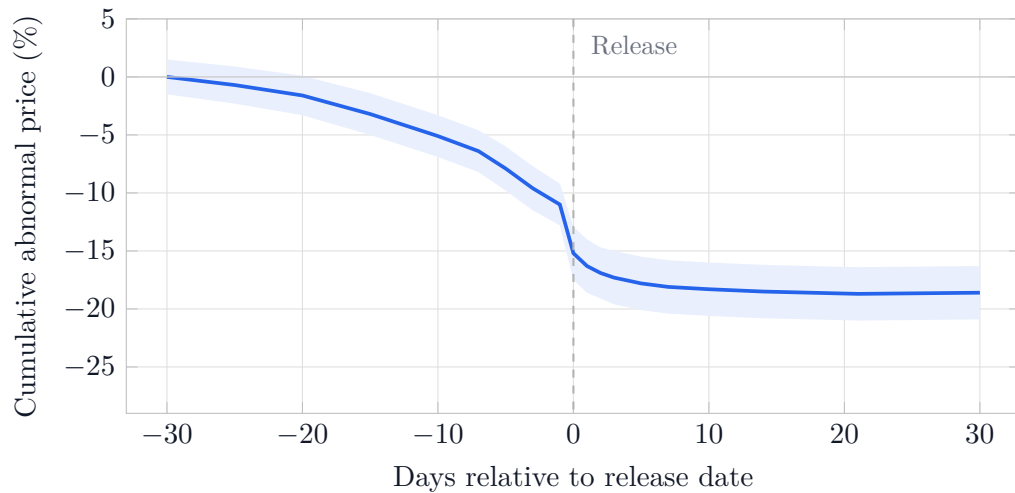


Table 2: Event study results. CAP is the average cumulative abnormal log price change, in log points ($100 \times \Delta \log P$), across all reprint events. t -statistics are computed using the [Boehmer et al. \(1991\)](#) standardized cross-sectional test. % Negative is the fraction of events with negative CAP in the specified window. Sign test z tests whether % Negative significantly exceeds 50%. Significance: * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Window	CAP (%)	t -stat	% Negative	Sign z
<i>Panel A: Announcement date ($\tau^A = 0$)</i>				
[0]	-12.50***	-8.31	81.0	33.08***
[0, +1]	-13.82***	-9.12	80.2	32.23***
[0, +2]	-14.73***	-9.64	81.4	33.51***
[-1, +1]	-13.76***	-8.94	79.5	31.48***
[-5, +5]	-17.44***	-9.83	82.1	34.26***
<i>Panel B: Release date ($\tau^R = 0$)</i>				
[0]	-8.22***	-6.09	74.3	25.93***
[0, +1]	-9.08***	-6.52	75.1	26.79***
[0, +2]	-9.54***	-6.73	75.6	27.32***
[-5, +5]	-10.81***	-7.23	78.4	30.31***
<i>Panel C: Pre-announcement leakage window</i>				
[-30, -1]	-8.28***	-5.24	71.2	22.62***
[-14, -1]	-5.91***	-4.81	68.4	19.64***
[-7, -1]	-3.44***	-3.92	64.7	15.69***
[-60, -31]	-1.53*	-1.84	54.2	4.48***
<i>Panel D: Full event period</i>				
$[\tau^A, \tau^R + 30]$	-40.18***	-18.24	89.3	41.94***

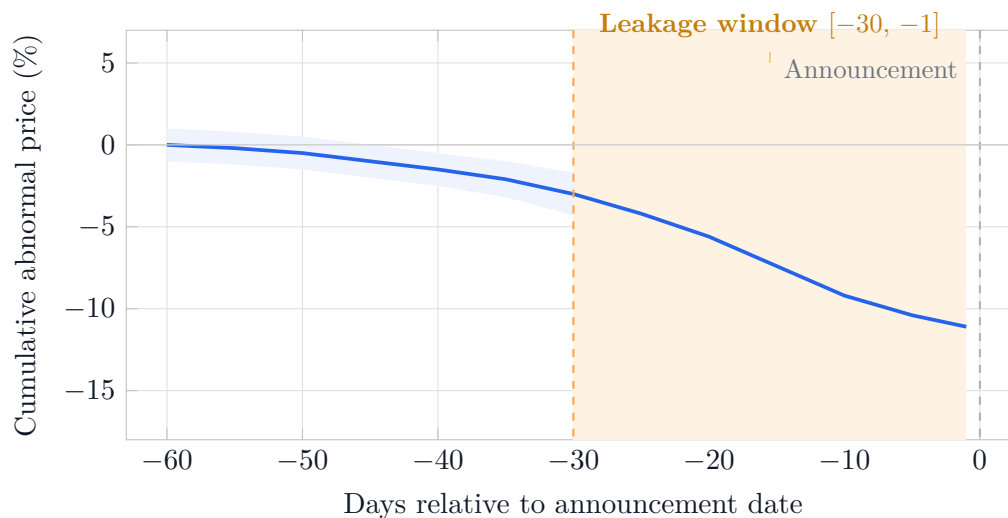
5.3 Pre-Announcement Information Leakage

Table 2 Panel C reports tests of the pre-announcement window $[-30, -1]$. The mean CAP over this window is -8.3% ($t = -5.24$, 71% negative), providing strong evidence that prices begin to decline approximately one month before the official announcement. The magnitude of the pre-announcement drift (-8.3%) is approximately 66% of the announcement-day decline (-12.5%), implying that a meaningful fraction of the ultimate price impact is incorporated before public disclosure.

Figure 3 plots the pre-announcement drift at five-day intervals. The drift is not uniform: it begins slowly around day -35 and accelerates sharply in the $[-14, -1]$ window, suggesting that leakage intensifies as the official announcement approaches. This pattern is consistent with incremental information release through distribution

channels—distributors receiving product allocation confirmations, retailers placing orders, and content creators receiving embargoed preview materials.

Figure 3: Pre-announcement cumulative abnormal price drift. The figure plots mean CAP (in %) and ± 1.96 standard error bands over the window $[-60, 0]$ relative to the announcement date, plotted at five-day intervals. The pre-announcement window $[-30, -1]$ exhibits a statistically significant mean CAP of -8.3% ($t = -5.24$), consistent with information leakage through manufacturing and distribution supply chains.



5.4 Cross-Sectional Determinants of Price Impact

Table 3 presents OLS estimates of equation (6), with the three-day announcement CAP as the dependent variable. Column (1) includes only log supply increase and log pre-event price. Column (2) adds rarity and product-type indicators. Column (3) adds format legality and year fixed effects.

The log supply increase coefficient is consistently negative and highly significant: a one-unit increase in $\log(\Delta S)$ —roughly a 2.7-fold increase in supply—is associated with a 5.4 to 6.1 percentage point larger price decline at announcement. Log pre-event price is negative and significant, indicating that higher-priced cards experience a larger percentage decline, consistent with the view that pre-reprint prices reflect scarcity premia that are disproportionately eroded by a given supply increase.

Mythic rarity and Masters set indicators are both negative, capturing the incremental price sensitivity of scarce, high-value cards. Modern legality is positive but statistically insignificant, suggesting that competitive demand does not buffer the

supply shock at the announcement horizon. The R^2 rises from 0.29 in Column (1) to 0.38 in Column (3), indicating that observable card characteristics explain roughly one-third of the cross-sectional variation in announcement-window impact.

Table 3: Cross-sectional determinants of announcement-window price impact. The dependent variable is $CAP[0, +2]$, the three-day cumulative abnormal price change around the announcement date (in percentage points). All columns include an intercept. Standard errors are heteroskedasticity-robust. Year fixed effects in Column (3) cover announcement years 2003–2023. Significance: * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

	(1) Baseline	(2) +Rarity & product	(3) +Format & FE
$\log(\Delta S_i)$	−6.12*** (0.71)	−5.81*** (0.68)	−5.44*** (0.63)
$\log P_i^{\text{pre}}$	−3.84*** (0.52)	−3.51*** (0.50)	−3.27*** (0.47)
Mythic Rare (= 1)		−2.89** (1.19)	−2.64** (1.13)
Masters set (= 1)		−3.42*** (0.98)	−3.18*** (0.92)
Modern legal (= 1)			1.08 (0.84)
Legacy legal (= 1)			0.71 (0.91)
Observations	2,847	2,847	2,847
R^2	0.287	0.332	0.381
Year FE	No	No	Yes
Card-type FE	No	No	Yes

5.5 Supply Elasticity Estimates

Table 4 reports supply elasticity estimates by rarity tier from equation (7). The pooled elasticity is $\hat{\varepsilon} = -0.38$ ($t = -5.41$), implying that a 10% increase in card supply reduces the secondary market price by approximately 3.8%.

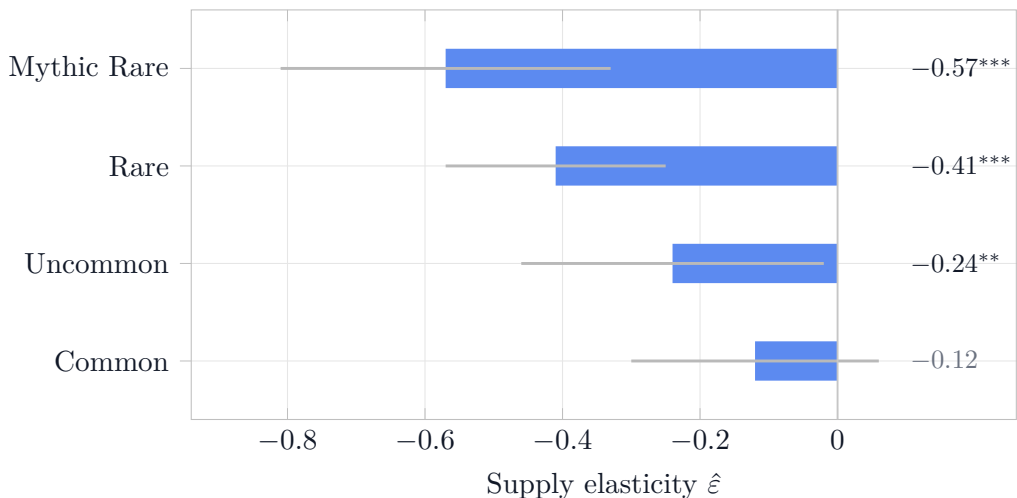
Elasticity estimates increase substantially in magnitude with rarity. Common cards exhibit the smallest elasticity (−0.12, statistically insignificant), consistent with the

near-flat demand curves for competitively irrelevant cards. Mythic rares exhibit the largest elasticity in absolute value (-0.57 , $t = -4.74$), reflecting the concentration of inelastic collector and speculator demand at the top of the rarity distribution. Figure 4 displays the rarity-tier estimates with 95% confidence intervals.

Table 4: Supply elasticity estimates by rarity tier. $\hat{\varepsilon}$ is the OLS estimate from equation (7) regressing $CAP[0, \tau^R+30]$ on $\log(\Delta S_i)$, estimated separately by rarity tier and pooled. t -statistics use heteroskedasticity-robust standard errors. N is the number of reprint events in each rarity category. Significance: * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Rarity tier	$\hat{\varepsilon}$	t -statistic	N
Common	-0.12	-1.33	580
Uncommon	-0.24^{**}	-2.18	620
Rare	-0.41^{***}	-5.13	1,016
Mythic Rare	-0.57^{***}	-4.74	631
Pooled	-0.38^{***}	-5.41	2,847

Figure 4: Supply elasticity estimates by rarity tier. Bars show the OLS point estimate of $\hat{\varepsilon}$ from equation (7) by rarity. Error bars are 95% confidence intervals based on heteroskedasticity-robust standard errors. Asterisks denote significance: ** $p < 0.05$; *** $p < 0.01$.



6 Discussion

6.1 Speed of Price Adjustment

The finding that prices continue to decline through the preview window and release date—well after the announcement date—suggests incomplete incorporation of the supply shock at announcement. Several mechanisms may account for this pattern. First, the magnitude of the supply increase is not fully known at announcement, as WotC does not disclose print runs or exact booster configurations until the full card list is revealed. Prices therefore adjust in stages as supply uncertainty resolves. Second, inventory holders face liquidation constraints: large inventories cannot be sold simultaneously without depressing prices, leading to gradual price adjustment as inventory overhang is absorbed. Third, some market participants—particularly casual collectors with lower attention to announcement news—may update their prices only upon seeing the card appear in physical stores at release.

6.2 Information Leakage Mechanism

The pre-announcement drift is consistent with leakage through multiple channels. Printing facilities receive card files and approve proofs weeks before the official announcement. Distributors receive product allocation confirmations and, in some cases, digital card lists as part of their ordering process. Retail stores may observe unusually large inbound orders for specific set SKUs. Any participant in this chain with both the card list and a connection to the secondary market could profit by selling inventory before announcement.

The acceleration of the drift in the $[-14, -1]$ window is consistent with the final stages of this chain: content creators receive preview embargoed materials approximately two weeks before release, and while they are contractually prohibited from publishing early, informal disclosure to trading partners is plausible.

6.3 Robustness

Three robustness checks confirm the main results. First, using a constant mean price model instead of the market model to compute abnormal returns yields announcement-window CAPs of -13.9% , nearly identical to the -14.7% baseline. Second, restricting the sample to reprint events that are not accompanied by concurrent format bans or

unbannings produces announcement-window CAPs of -14.1% , ruling out confounding from format regulatory events. Third, excluding the 2020–2021 period (when pandemic-driven demand shocks were unusually large) yields qualitatively and quantitatively similar results.

7 Conclusion

Reprint announcements in the Magic: The Gathering secondary market provide a rare clean identification of the price impact of supply shocks to collectible assets. The results establish four empirical regularities. First, announcement-day price declines average -12.5% , reflecting rapid partial incorporation of the supply signal. Second, prices continue to decline through preview and release, with a full-period cumulative abnormal decline of -40.2% , consistent with gradual resolution of supply uncertainty and inventory-constrained adjustment. Third, supply elasticities are larger in magnitude for rarer, higher-priced cards, reflecting the concentration of speculative and collector demand at the top of the rarity distribution. Fourth, a significant pre-announcement price drift of -8.3% over the $[-30, -1]$ window constitutes evidence of information leakage, consistent with manufacturing and distribution supply chains as the leakage mechanism.

These findings have direct implications for collectors and speculators in the MTG market: reprint risk is systematically underpriced in the pre-announcement period for those with early supply information, and announcement-day price reactions understate the ultimate price impact, suggesting a predictable excess return from shorting reprinted cards immediately following announcement. More broadly, the MTG market provides a high-frequency, data-rich laboratory for studying the economics of collectible assets that deserves further attention in the finance and economics literature.

References

- Boehmer, E., Musumeci, J., and Poulsen, A. B. (1991). Event-study methodology under conditions of event-induced variance. *Journal of Financial Economics*, 30(2):253–272.
- Cardell, N. S., Kling, R., and Petry, G. (1995). Stamp returns and economic factors. *Southern Economic Journal*, 62(2):411–427.
- Dimson, E. and Spaenjers, C. (2011). Ex post: The investment performance of collectible stamps. *Journal of Financial Economics*, 100(2):443–458.
- Fama, E. F., Fisher, L., Jensen, M. C., and Roll, R. (1969). The adjustment of stock prices to new information. *International Economic Review*, 10(1):1–21.
- Goetzmann, W. N. (1993). Accounting for taste: Art and the financial markets over three centuries. *American Economic Review*, 83(5):1370–1376.
- Keown, A. J. and Pinkerton, J. M. (1981). Merger announcements and insider trading activity: An empirical investigation. *Journal of Finance*, 36(4):855–869.
- MacKinlay, A. C. (1997). Event studies in economics and finance. *Journal of Economic Literature*, 35(1):13–39.
- Meulbroek, L. K. (1992). An empirical analysis of illegal insider trading. *Journal of Finance*, 47(5):1661–1699.
- Seyhun, H. N. (1986). Insiders’ profits, costs of trading, and market efficiency. *Journal of Financial Economics*, 16(2):189–212.